Joy **Dhar**

DEEP LEARNING RESEARCHER | PH.D. SCHOLAR IN COMPUTER SCIENCE & ENGINEERING · IIT ROPAR

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"Building effective and reliable AI for those who need it most—not just where data is abundant, but also where care is scarce."

Summary

Ph.D. scholar and advanced deep-learning researcher at IIT Ropar, specializing in heterogeneous medical data—medical imaging, multi-omics, electronic health records, and time-series data—to develop robust and efficient AI-driven healthcare applications. My work focuses on multimodal fusion, adversarially robust neural architectures, and computationally efficient model design, with publications in top-tier venues. I aim to translate cutting-edge research into reliable decision-support AI systems for resource-constrained environments.

Research Interests

Multimodal & Generative Models: Multimodal Fusion; Multi & Cross-Modal Attention; Transformers; Visual–Language Models & GANs.

Robustness & Reliability Analysis: Adversarial Attacks; Certified & Empirical Defenses; Uncertainty Quantification & Curriculum Learning.

Domain Adaptation Analysis: Knowledge Distillation; Modality Agnostic & Domain Generalization.

Foundations & Architectures: Supervised; Self-Supervised; Unsupervised; Continual; Evidence; Multitask & Physics-Inspired Learning.

Biomedical & Healthcare AI: Cost-Effective Medical Image Analysis; EHR & Time-Series Analysis; Multi-Disease Classification & Segmentation.

Publications

Conference

1 **J. Dhar**, S. Xia, M. Pandey, M. Haghighat, A. Alavi, F. Sohel, W. Zhang, and N. Zaidi, "Certified VS. Empirical adversarial robustness via hybrid convolutions with attention stochasticity," ***ICLR 2026***, (**Accepted**).

2 **J. Dhar**, N. Zaidi, M. Haghighat, "Effective and Robust Multimodal Medical Image Analysis," in Proc. of the ***ACM SIGKDD*** *Conference on Knowledge Discovery and Data Mining*, South Korea, 2026, (**Accepted**).

3 **J. Dhar**, M. Pandey, D. Chakladar, M. Haghighat, A. Alavi, S. Mistry, N. Zaidi, "HyPCA-Net: Advancing Multimodal Fusion in Medical Image Analysis," in *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision* (**WACV**), Tucson, AZ, USA, Feb. 2026, (**Accepted**).

4 **J. Dhar**, N. Zaidi, M. Haghighat, S. Roy, P. Goyal, A. Alavi, and V. Kumar, "Multimodal fusion learning with dual attention for medical imaging," in *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision* (**WACV**), Tucson, AZ, USA, Feb. 2025, pp. 4362–4371. doi: 10.1109/WACV61041.2025.00428

5 **J. Dhar**, K. Rana, and P. Goyal, "Uncertainty-aware robust information fusion attention network for brain tumors classification in MRI images," in *Proc. 2024 Int. Conf. on Pattern Recognition* (**ICPR**), Kolkata, India, Dec. 2024, pp. 311–327. doi: 10.1007/978-3-031-78398-2_21

6 M. S. Kanroo, H. S. Kawoosa, **J. Dhar**, and P. Goyal, "PiExtract: an end-to-end data extraction pipeline for pie charts," in *Proc. 2024 Int. Conf. on Pattern Recognition* (**ICPR**), Kolkata, India, Dec. 2024, pp. 31–46. doi: 10.1007/978-3-031-78122-3_3

Journal

1 **J. Dhar**, P. Goyal, M. Haghighat, N. Zaidi, F. Sohel, Q. B. Vo, and K. C. Santosh, "Towards building robust models for unimodal and multimodal medical imaging data," ***Information Fusion***, vol. 127, pp. 103822, March 2026. doi: 10.1016/j.inffus.2025.103822.

2 **J. Dhar**, K. Rana, P. Goyal, A. Alavi, R. Rana, Q. B. Vo, S. Mishra, and S. Mistry, "Dual-phase neural networks for feature extraction and ensemble learning for recognizing human health activities," ***Applied Soft Computing***, vol. 169, pp. 112550, Jan. 2025. doi: 10.1016/j.asoc.2024.112550.

3 **J. Dhar** and S. Roy, "Identification and diagnosis of cervical cancer using a hybrid feature selection approach with the Bayesian optimization–based optimized CatBoost classification algorithm," *J. Ambient Intell. Humaniz. Comput.*, vol. 15, pp. 3459–3477, Jun. 2024. doi: 10.1007/s12652-024-04825-8.

4 **J. Dhar**, "An adaptive intelligent diagnostic system to predict early stage of Parkinson's disease using two-stage dimension reduction with genetically optimized LightGBM algorithm," *Neural Computing & Appllications*, vol. 34, no. 6, pp. 4567–4593, Jun. 2022. doi: 10.1007/s00521-021-06612-4.

5 **J. Dhar** and N. A. Ayele, "Multi-tier ensemble learning model with neighborhood component analysis to predict health diseases," *IEEE Access*, vol. 9, pp. 138677–138715, 2021. doi: 10.1109/ACCESS.2021.3117963.

6 **J. Dhar**, "Multistage ensemble learning model with weighted voting and genetic algorithm optimization strategy for detecting chronic obstructive pulmonary disease," *IEEE Access*, vol. 9, pp. 48640–48657, 2021. doi: 10.1109/ACCESS.2021.3067949.

7 **J. Dhar** and A. K. Jodder, "An effective recommendation system to forecast the best educational program using machine learning classification algorithms," *Ingénierie Systèmes d'Information*, vol. 25, no. 5, pp. 559–568, Nov. 2020. doi: 10.18280/isi.250502.

Manuscripts Under Review

Conference Submissions

1 **J. Dhar**, M. Pandey, B. Bozorgtabar, N. Zaidi, W. Zhang, W. H. Li, T. Mu, D. Mahapatra, M. Baktashmotlagh, T. Le, C. Chen, S. Mistry, C. Gonzalez, S. E. Kahou, L. Yao, P. Koniusz, R.B. Fisher, D. Phung, B. Han, N. Vasconcelos, and P. Lio, "Does a Hybrid Space-Aware Randomized Defense Improve Empirical and Certified Adversarial Robustness?," submitted to ***ICML 2026***, manuscript under review.

2 **J. Dhar**, M. Pandey, N. Zaidi, W. H. Li, W. Zhang, B. Bozorgtabar, D. Mahapatra, T. Mu, C. Chen, C. Gonzalez, T. V. Nguyen, M. Yang, C. Sun, E. Yang, N. Aghaeepour, M. Baktashmotlagh, D. Phung, R.B. Fisher, C. Davatzikos, "DEMAnD: Redefining Decoder Design to Advance Medical Image Segmentation," submitted to ***KDD 2026***, manuscript under review.

3 **J. Dhar**, C. Chen, N. Zaidi, M. Haghighat, P. Goyal, M. Pandey, and F. Sohel, "Advancing Multimodal Fusion on Heterogeneous Medical Data with Hybrid Geometry Attention," submitted to ***KDD 2026***, manuscript under review.

4 D. Mahapatra, A. Das, M. Pandey, **J. Dhar**, S. Roy, Z. Ge, B. Bozorgtabar, and I. Razzak, "Proximity-Constrained Counterfactual Decoding for Hallucination-Robust Medical VQA," submitted to ***ECCV 2026***, manuscript under review.

5 **J. Dhar**, M. Pandey, D. Mahapatra, P. Koniusz, N. Zaidi, I. Razzak, B. Bozorgtabar, C. Sun, J. Yang, M. Reyes, Z. Ge, L. Yao, M. Baktashmotlagh, A. Farshad, I. Banerjee, W. Zhang, M. Fiaz, C. Gonzalez, R. Narava, A. Shekhar, V. Gupta, E. Yang, H. Müller, K. Åström, N. Vasconcelos, P. Lio, "Can Multimodal Fusion of Heterogeneous Medical Data Accelerate Convergence?," submitted to ***ECCV 2026***, manuscript under review.

6 **J. Dhar**, D. Chakladar, M. Pandey, D. Mahapatra, N. Zaidi, B. Bozorgtabar, I. Razzak, H. Mokayed, P. Goyal, "GCARE: Advancing Multimodal Fusion on Heterogeneous Medical Data," submitted to ***ECCV 2026***, manuscript under review.

7 **J. Dhar**, M. Pandey, P. Goyal, B. Bozorgtabar, P. Liu, D. Mahapatra, and P. Lio, "CoVANT: Advancing Multimodal Fusion on Heterogeneous Medical Data," submitted to ***NeurIPS 2026***, manuscript under review.

8 **J. Dhar**, M. Pandey, P. Goyal, D. Mahapatra, and P. Lio, "GSAND: Hybrid-Space Attention Advances Medical Image Segmentation," submitted to ***NeurIPS 2026***, manuscript under review.

9 **J. Dhar**, M. Pandey, P. Goyal, D. Mahapatra, and P. Lio, "Certified vs. Empirical Adversarial Robustness in Multimodal Fusion via Hybrid Space Convolutional Stochasticity," submitted to ***NeurIPS 2026***, manuscript under review.

Journal Submissions

1 **J. Dhar**, P. Goyal, and F. Sohel, "MANAT: Multimodal attention noise injection with adversarial training for adversarial robustness in medical image analysis," submitted to *Applied Soft Computing*, manuscript under review.

Education

Indian Institute of Technology Ropar (IIT Ropar) *Punjab, India*
PH.D. IN COMPUTER SCIENCE AND ENGINEERING *Jul. 2022––Jun. 2027*
- Advisor: Dr. Puneet Goyal; Institute Scholarship Recipient

Maulana Abul Kalam Azad University of Technology *West Bengal, India*
M.TECH. IN COMPUTER SCIENCE AND ENGINEERING *Aug. 2015––Nov. 2017*
- CGPA 8.92/10 (A+); ranked 3rd in the department

West Bengal University of Technology *West Bengal, India*
B.TECH. IN COMPUTER SCIENCE AND ENGINEERING *Aug. 2009––Nov. 2012*
- CGPA 8.59/10 (A+); ranked 10th in the department

Awards & Honors

2025	**Recipient**, Anusandhan National Research Foundation (ANRF) Travel Fund to present at WACV 2025	*Tucson, USA*
2024	**Winner**, IEEE Video and Image Processing (VIP) Cup at ICIP	*Abu Dhabi, UAE*
2011	**Finalist**, IBM India The Great Mind Challenge (TGMC) Project	*Bangalore, India*

Scholarships/Fellowships

INTERNATIONAL

2022	**Deakin University Postgraduate Research Scholarship**, Deakin University	*Australia*
2022	**Industrial Scholarship for Ph.D., Big Data and High-Performance Computing**, University of Genoa	*Italy*
2022	**Ph.D. Institute Scholarship**, University of Salerno	*Italy*
2022	**JCU Postgraduate Research Stipend Scholarship**, James Cook University	*Australia*
2021	**Ph.D. Institute Scholarship**, National Taipei University of Technology	*Taiwan*
2021	**Ph.D. Institute Scholarship**, National Dong Hwa University	*Taiwan*
2021	**Elite Doctoral Scholarship**, National Sun Yat-sen University	*Taiwan*

Domestic

2022	**Institute Scholarship**, Indian Institute of Technology Dhanbad	*India*
2022	**Institute Scholarship**, Indian Institute of Technology Ropar	*India*
2019	**NET Fellowship**, NTA-UGC	*India*
2018	**GATE Fellowship**, M.H.R.D.	*India*

Current Projects

Does a Hybrid Space-Aware Weight Refiner Improve Multimodal Fusion Robustness? *Punjab, India*

DEVISING HYBRID SPACE AWARE ATTENTION FOR MULTIMODAL FUSION *Feb. 2026*

- Devised a hybrid space aware weight refinement module that incorporates both implicit and explicit refinement process to improve multimodal fusion robustness.

Robust-Zero: A Defense Framework for Zero-Shot Classification *Punjab, India*

DEVELOPING A ZERO-SHOT DEFENSE FRAMEWORK USING RANDOM FILTERS, ATTENTION NOISE, AND FEATURE DISENTANGLEMENT *May 2025*

- Jointly exploit random filtering, attention-noise injection, and disentangled feature representations to mitigate vulnerabilities.

Academic Services

2021–2026	**PC member/Reviewer**, CVPR; AAAI; KDD; ACM MM; WACV; ICPR; ASOC; NCAA; Scientific Reports	*India & Taiwan*
2022–2027	**Teaching Assistant**, Indian Institute of Technology Ropar	*Punjab, India*
2021–2022	**Research Assistant**, National Sun Yat-sen University	*Kaohsiung, Taiwan*
2016–2021	**Information Technology Teacher**, Hatgobindapur M. C. High School	*Burdwan, India*
2014–2016	**Lecturer in Computer Science & Technology**, JLD College of Engineering & Management	*Kolkata, India*

Technical Skills

Programming: Python, C/C++, Java, C#, PHP, MATLAB;

Frameworks: TensorFlow/Keras, PyTorch, scikit-learn;

Data Analysis: NumPy, Pandas, OpenCV, SciPy;

Tools & Platforms: LaTeX, Jupyter, Kaggle, Google Colab, VS Code.

Selected Coursework

Deep Learning	Computer Vision	Advanced Computer Vision	Digital Image Processing
Internet of Things	Pattern Recognition	Artificial Intelligence	Data Structures & Algorithms

References

Dr. Puneet Goyal
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